

PADHAISHURU

GATE Preparation Platform

GATE EC

Electronics & Communication Engineering

TREND-BASED MOCK PAPER

— 03 —

Duration	3 Hours
Total Questions	65
Maximum Marks	100
Sections	9 subjects

Difficulty Distribution

Easy: 22 questions (34%)

Moderate: 32 questions (49%)

Difficult: 11 questions (17%)

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
These mock papers are constructed using analysis of historical GATE
PYQ patterns.
Historical patterns do not guarantee future examination questions.

GENERAL INSTRUCTIONS

GATE EC — Trend-Based Mock Paper

1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
 - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
 - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
 - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
 - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
 - NAT questions carry no negative marking.
 - MSQ questions carry no negative marking.
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4. How to Answer

- For MCQ: Select exactly one option.
 - For MSQ: Select one or more options. Partial credit is NOT awarded.
 - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
 - The Question Booklet contains multiple pages. Verify the page count.
 - Immediately report any discrepancy in the Question Booklet to the invigilator.
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Q1.

2M [no -ve]

[General Aptitude]

Five years ago, the ratio of Aman's age to his father's age was 1:4, and five years from now, the ratio will be 2:5. What was his father's age when Aman was born?

- (A) 28 years
- (B) 30 years
- (C) 35 years
- (D) 32 years

Q2.

2M [-0.66]

[General Aptitude]

Here are two analogous groups, Group-I and Group-II, that list words in their decreasing order of intensity. Identify the missing word in Group-II. Group-I: Abuse Insult Ridicule Group-II: _____ Praise Appreciate

- (A) Extol
- (B) Prize
- (C) Appropriate
- (D) Espouse

Q3.

2M [-0.66]

[General Aptitude]

Sum of first 100 natural numbers:

- (A) 1000
- (B) 5050
- (C) 4950
- (D) 5000

Q4.

3M [-0.66]

[General Aptitude]

The greatest prime factor of (3199 - 3196) is

- (A) 13
- (B) 17
- (C) 3
- (D) 11

Q5.

2M [no -ve]

[General Aptitude]

If train 100m crosses pole in 10s, speed:

- (A) 10 m/s
- (B) 36 km/h
- (C) 18 km/h
- (D) 72 km/h

Q6.

2M [no -ve]

[General Aptitude]

Had I learnt acting as a child, I _____ a famous film star. Select the most appropriate option to complete the above sentence.

Enter your answer here

Q7.

2M [no -ve]

[General Aptitude]

If ' ' denotes increasing order of intensity, then the meaning of the words [charm enamor bewitch] is analogous to [bored _____ weary]. Which one of the given options is appropriate to fill the blank?

Enter your answer here

Q8.

2M [-0.66]

[General Aptitude]

Choose opposite of 'benevolent':

- (A) Hostile
- (B) Kind
- (C) Generous
- (D) Helpful

Q9.

2M [no -ve]

[General Aptitude]

The following figures show three curves generated using an iterative algorithm. The total length of the curve generated after 'Iteration ' is: Note: The figures shown are representative. 1 Iteration 0: Iteration 1: 1
1 Length of each segment: 3 3 1 1 Iteration 2: 9 Length of each segment: 9

Enter your answer here

Q10.

2M [no -ve]

[General Aptitude]

P, Q, R, S, and T have launched a new startup. Two of them are siblings. The office of the startup has just three rooms. All of them agree that the siblings should not share the same room. If S and Q are single children, and the room allocations shown below are acceptable to all, PR TS Q PQ RT S then, which one of the given options is the siblings?

Enter your answer here

Q11.

2M [-0.66]

[Engineering Mathematics]

Integral of $1/x$ dx:

- (A) x
- (B) $\log(x)+C$
- (C) $1/x^2$
- (D) $x^{2/2}$

Q12.

2M [-0.66]

[Engineering Mathematics]

$i^4 =$:

- (A) 0
- (B) 1
- (C) -1
- (D) i

Q13.

2M [no -ve]

[Engineering Mathematics] $P(A \cap B) = ?$

- (A) $P(A) + P(B)$
- (B) $P(A) + P(B) - P(A \cap B)$
- (C) $P(A)P(B)$
- (D) $P(A)/P(B)$

Q14.

3M [-0.66]

[Network, Signals & Systems]

The photocurrent of a PN junction diode solar cell is 1 mA. The voltage corresponding to its maximum power point is 0.3 V. If the thermal voltage is 30 mV, the reverse saturation current of the diode (in nA, rounded off to two decimal places) is _____.

- (A) A
- (B) B
- (C) C
- (D) D

Q15.

3M [-0.66]

[Network, Signals & Systems]

Consider a system represented by the block diagram shown below. Which of the following signal flow graphs represent(s) this system? Choose the correct option(s).

- (A) A
- (B) B
- (C) C
- (D) D

Q16.

3M [-0.66]

[Network, Signals & Systems]

and are Bernoulli random variables taking values in $\{0,1\}$. The joint probability mass function of the random variables is given by: $(= 0, = 0) = 0.06$ $(= 0, = 1) = 0.14$ $(= 1, = 0) = 0.24$ $(= 1, = 1) = 0.56$ The mutual information ($I(X; Y)$) is _____ (rounded off to two decimal places). Electronics and Communication Engineering (EC)

- (A) A
- (B) B
- (C) C
- (D) D

Q17.

1M [-0.33]

[Network, Signals & Systems]

Convolution in time =

- (A) Addition in freq
- (B) Multiplication in freq
- (C) Division
- (D) None

Q18.

2M [-0.66]

[Network, Signals & Systems]

Consider a MOS capacitor made with p-type silicon. It has an oxide thickness of 100 nm, a fixed positive oxide charge of 10^{-8} C/cm^2 at the oxide-silicon interface, and a metal work function of 4.6 eV. Assume that the relative permittivity of the oxide is 4 and the absolute permittivity of free space is $8.85 \times 10^{-14} \text{ F/cm}$. If the flatband voltage is 0 V, the work function of the p-type silicon (in eV, rounded off to two decimal places) is _____.

- (A) A
- (B) B
- (C) C
- (D) D

Q19.

2M [no -ve]

[Network, Signals & Systems]

Let $X(t) = A \cos(2\pi f_0 t + \phi)$ be a random process, where amplitude A and phase ϕ are independent of each other, and are uniformly distributed in the intervals $[-2, 2]$ and $[0, 2\pi]$, respectively. $X(t)$ is fed to an 8-bit uniform mid-rise type quantizer. Given that the autocorrelation of $X(t)$ is $R_X(\tau) = 2 \cos(2\pi f_0 \tau)$, the signal to 3 quantization noise ratio (in dB, rounded off to two decimal places) at the output of the quantizer is _____.

Enter your answer here

Q20.

2M [no -ve]

[Network, Signals & Systems]

Which of the following statements is/are true for a BJT with respect to its DC current gain?

Enter your answer here

Q21.

2M [no -ve]

[Network, Signals & Systems]

A full adder and an XOR gate are used to design a digital circuit with inputs X , Y , and Z , and output F , as shown below. The input Z is connected to the carry-in input of the full adder. If the input Z is set to logic '1', then the circuit functions as _____ with X and Y as inputs.

Enter your answer here

Q22.

1M [no -ve]

[Electronic Devices]

SCR has:

Enter your answer here

Q23.

1M [-0.33]

[Electronic Devices]

Zener diode used for:

- (A) Rectification
- (B) Voltage regulation
- (C) Amplification
- (D) None

Q24.

1M [-0.33]

[Electronic Devices]

n-type majority carriers:

- (A) Holes
- (B) Electrons
- (C) Both equal
- (D) None

Q25.

1M [-0.33]

[Electronic Devices]

MOSFET is:

- (A) BJT
- (B) Unipolar
- (C) Bipolar
- (D) None

Q26.

1M [-0.33]

[Analog Circuits]

RC oscillator frequency:

- (A) $f = 1/(2\pi RC)$
- (B) $f = 1/RC$
- (C) $f = 1/(4\pi RC)$
- (D) $f = 2\pi RC$

Q27.

1M [no -ve]

[Analog Circuits]

;" —â \$¥B —3

Enter your answer here

Q28.

1M [-0.33]

[Analog Circuits]

FET is:

- (A) Current controlled
- (B) Voltage controlled
- (C) Power
- (D) None

Q29.

1M [-0.33]

[Analog Circuits]

Ideal op-amp open-loop gain:

- (A) 0
- (B) 1
- (C) Infinite
- (D) -1

Q30.

1M [no -ve]

[Analog Circuits]

Common emitter has:

*Enter your answer here***Q31.**

1M [no -ve]

[Digital Circuits]

D flip-flop follows input after:

*Enter your answer here***Q32.**

1M [-0.33]

[Digital Circuits]

4-bit binary counter counts:

- (A) 0-15
- (B) 0-9
- (C) 1-16
- (D) 1-10

Q33.

1M [-0.33]

[Digital Circuits]

ROM is:

- (A) Read only
- (B) Read write
- (C) Volatile
- (D) Cache

Q34.

1M [-0.33]

[Digital Circuits]

TTL stands for:

- (A) Transistor-Transistor Logic
- (B) Tunnel Transistor Logic
- (C) Totem Pole Logic
- (D) None

Q35.

1M [-0.33]

[Control Systems]

Transfer function is defined:

- (A) In time domain
- (B) Laplace
- (C) Fourier
- (D) Z-domain

Q36.

1M [-0.33]

[Control Systems]

Routh-Hurwitz determines:

- (A) Stability
- (B) Time response
- (C) Frequency
- (D) Sensitivity

Q37.

1M [-0.33]

[Control Systems]

Lead compensator:

- (A) Improves steady state
- (B) Improves transient
- (C) Both
- (D) None

Q38.

1M [-0.33]

[Control Systems]

Nyquist criterion uses:

- (A) Polar plot
- (B) Root locus
- (C) Routh
- (D) Bode

Q39.

1M [-0.33]

[Communication Systems]

ASK modulation varies:

- (A) Amplitude
- (B) Frequency
- (C) Phase
- (D) None

Q40.

1M [no -ve]

[Communication Systems]

PCM stands for:

*Enter your answer here***Q41.**

1M [-0.33]

[Communication Systems]

FM has:

- (A) Constant amplitude
- (B) Variable
- (C) Both
- (D) None

Q42.

1M [-0.33]

[Communication Systems]

AM stands for:

- (A) Amplitude Modulation
- (B) Angle Modulation
- (C) Analog Modulation
- (D) None

Q43.

1M [-0.33]

[Electromagnetics]

Maxwell has how many equations:

- (A) 2
- (B) 3
- (C) 4
- (D) 5

Q44.

1M [-0.33]

*[Electromagnetics]*Characteristic impedance $Z_0 =$:

- (A) $\sqrt{L/C}$
- (B) L/C
- (C) C/L
- (D) $\sqrt{LC}$

Q45.

1M [-0.33]

[Electromagnetics]

Waveguide supports:

- (A) TEM
- (B) TE/TM
- (C) Both
- (D) None

Q46.

1M [-0.33]

[Electromagnetics]

Skin depth increases with:

- (A) Frequency
- (B) Conductivity
- (C) Both
- (D) Neither

Q47.

2M [no -ve]

[Network, Signals & Systems]

Consider a part of an electrical network as shown below. Some node voltages, and the current flowing through the 3 resistor are as indicated. The voltage (in Volts) at node is _____.

*Enter your answer here***Q48.**

1M [-0.33]

[Analog Circuits]

FET is:

- (A) Current controlled
- (B) Voltage controlled
- (C) Power
- (D) None

Q49.

1M [no -ve]

[Communication Systems]

PCM stands for:

Enter your answer here

Q50.

1M [-0.33]

[Digital Circuits]

TTL stands for:

- (A) Transistor-Transistor Logic
- (B) Tunnel Transistor Logic
- (C) Totem Pole Logic
- (D) None

Q51.

1M [no -ve]

[Network, Signals & Systems]

3-phase power:

Enter your answer here

Q52.

2M [no -ve]

[Network, Signals & Systems]

A source transmits symbol that takes values uniformly at random from the set $\{-2, 0, 2\}$. The receiver obtains $\hat{s} = s + n$, where n is a zero-mean Gaussian random variable independent of s . The receiver uses the maximum likelihood decoder to estimate the transmitted symbol $\hat{s}$. Suppose the probability of symbol estimation error is expressed as follows: $P_e = \alpha \beta$, where $\beta > 1$ denotes the probability that exceeds 1. What is the value of α ?

Enter your answer here

Q53.

2M [no -ve]

[Network, Signals & Systems]

Eight students (P, Q, R, S, T, U, V, and W) are playing musical chairs. The figure indicates their order of position at the start of the game. They play the game by moving forward in a circle in the clockwise direction. After the 1st round, 4th student behind P leaves the game. After 2nd round, 5th student behind Q leaves the game. After 3rd round, 3rd student behind V leaves the game. After 4th round, 4th student behind U leaves the game. Who all are left in the game after the 4th round? Note: The figure shown is representative. P W R U T V Q S

Enter your answer here

Q54.

2M [no -ve]

[Network, Signals & Systems]

Consider a frequency-modulated (FM) signal $s(t) = \cos(2\pi f_c t + 3 \sin(2\pi f_1 t) + 4 \sin(6\pi f_1 t))$, where A and f_c are, respectively, the amplitude and frequency (in Hz) of the carrier waveform. The frequency f_1 is in Hz, and assume that $f_1 > 1001$. The peak frequency deviation of the FM signal in Hz is _____.

Enter your answer here

Q55.

2M [no -ve]

[Network, Signals & Systems]

Let $\mathbb{R}$ and $\mathbb{C}$ denote the set of real numbers and the three dimensional vector space over it, respectively. The value of α for which the set of vectors $\{[2 \ -3], [3 \ -1 \ 3], [1 \ -5 \ 7]\}$ does not form a basis of $\mathbb{C}$ is _____.

Electronics and Communication Engineering (EC)

- (A) A
- (B) B
- (C) C
- (D) D

Q56.

2M [no -ve]

[Network, Signals & Systems]

A source transmits symbols from an alphabet of size 16. The value of maximum achievable entropy (in bits) is _____.

- (A) A
- (B) B
- (C) C
- (D) D

Q57.

2M [-0.66]

[Network, Signals & Systems]

In the network shown below, maximum power is to be transferred to the load Z_L . The value of R (in Ω) is _____. Electronics and Communication Engineering (EC)

- (A) A
- (B) B
- (C) C
- (D) D

Q58.

1M [-0.33]

[Engineering Mathematics]

$P(A \cap B) = ?$

- (A) $P(A) + P(B)$
- (B) $P(A) + P(B) - P(A \cap B)$
- (C) $P(A)P(B)$
- (D) $P(A)/P(B)$

Q59.

1M [no -ve]

[General Aptitude]

Sum of first 100 natural numbers:

Enter your answer here

Q60.

2M [no -ve]

[Network, Signals & Systems]

Let $x[n]$ be a discrete-time signal whose z -transform is $X(z)$. Which of the following statements is/are TRUE?

Enter your answer here

Q61.

2M [no -ve]

[Network, Signals & Systems]

In the circuit given below, the switch S was kept open for a sufficiently long time and is closed at time $t = 0$. The time constant (in seconds) of the circuit for $t > 0$ is _____. $t = 0$ 3H S 5A 2 4 4

Enter your answer here

Q62.

1M [-0.33]

[Network, Signals & Systems]

Impulse response of LTI gives:

- (A) Input
- (B) Output
- (C) Complete response
- (D) None

Q63.

2M [-0.66]

[Network, Signals & Systems]

A source transmits a symbol s , taken from $\{-4, 0, 4\}$ with equal probability, over an additive white Gaussian noise channel. The received noisy symbol r is given by $r = s + w$, where the noise w is zero mean with variance 4 and is independent of s . Using $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-t^2/2} dt$, the optimum symbol error probability is _____. 2 2

- (A) $2Q(2)$
- (B) $4Q(1)$
- (C) $2Q(1)$
- (D) $4Q(2)$

Q64.

2M [no -ve]

[Network, Signals & Systems]

In the circuit shown below, the AND gate has a propagation delay of 1 ns. The edge-triggered flip-flops have a set-up time of 2 ns, a hold-time of 0 ns, and a clock-to-Q delay of 2 ns. The maximum clock frequency (in MHz, rounded off to the nearest integer) such that there are no setup violations is _____.
Electronics and Communication Engineering (EC)

Enter your answer here

Q65.

1M [no -ve]

[General Aptitude]

The following figures show three curves generated using an iterative algorithm. The total length of the curve generated after 'Iteration' is: Note: The figures shown are representative. 1 Iteration 0: Iteration 1: 1
1 Length of each segment: 3 3 1 1 Iteration 2: 9 Length of each segment: 9

Enter your answer here